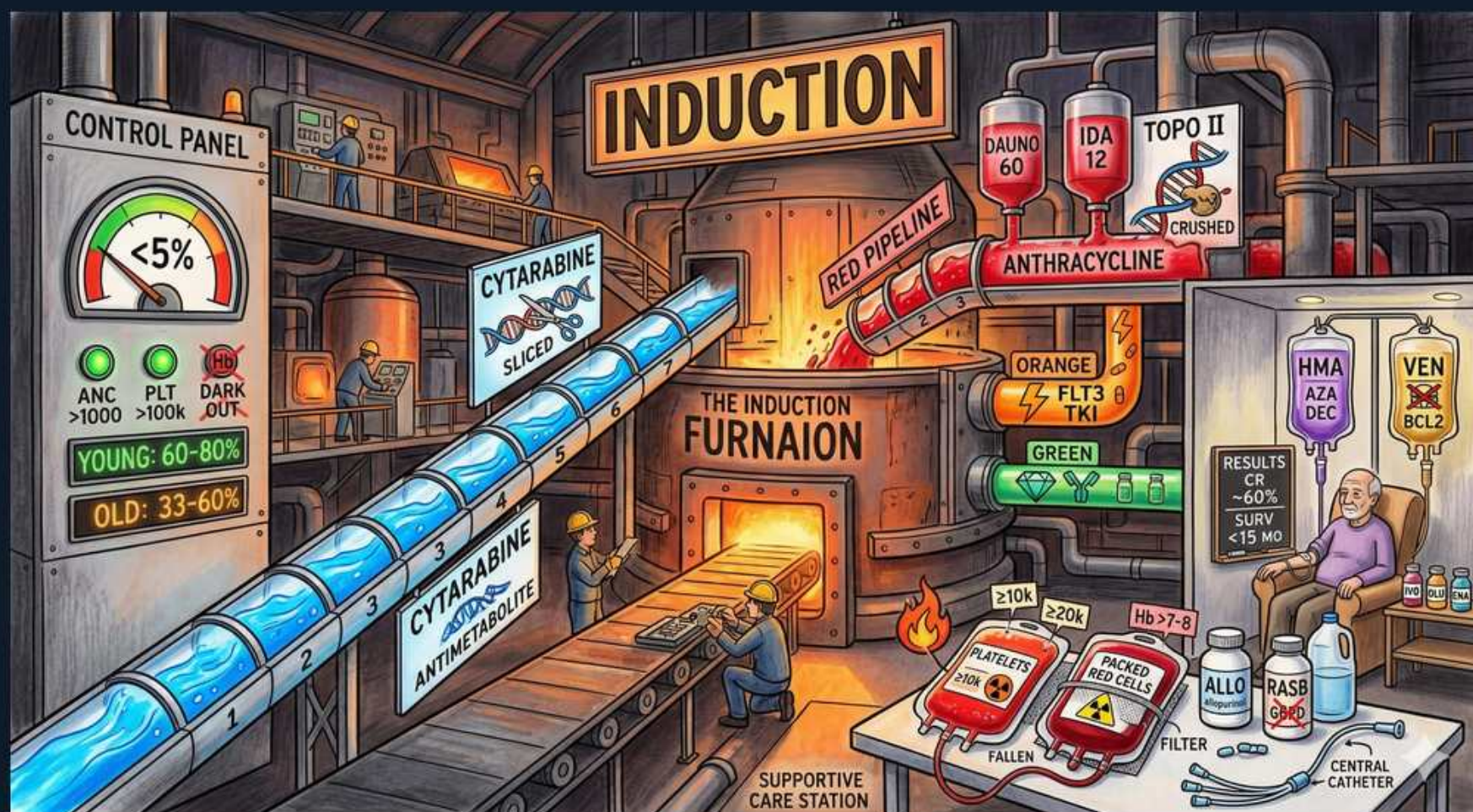


AML — Induction (7+3)



1. Induction furnace

WHAT YOU SEE

A blazing INDUCTION furnace dominating the center of the factory — the central fire is 7+3 induction whose goal is CR, not cure.

WHAT IT ENCODES

Induction is the first, intensive phase of AML therapy whose goal is complete remission (CR) — defined as <5% marrow blasts with hematologic recovery — not cure. CR is necessary but not sufficient: nearly all patients relapse without post-remission (consolidation) therapy, so induction is always followed by consolidation chemo and/or allo-HCT depending on ELN risk. Fit patients <60-75 yr with good performance status are candidates for intensive induction; APL (t(15;17)) is the exception and is treated with ATRA + arsenic trioxide, not 7+3.

BOARD TRAP

CR (<5% blasts) is the goal of induction, but achieving CR alone does NOT cure AML — consolidation is mandatory. Do not confuse induction with the curative endpoint.

2. Cytarabine 7 days

WHAT YOU SEE

A long cytarabine pipe/sled running across 7 segments — the 7-segment conveyor is the 7-day continuous cytarabine infusion of 7+3.

WHAT IT ENCODES

The 7+3 regimen pairs cytarabine (Ara-C) as a continuous IV infusion for 7 days with an anthracycline given on the first 3 days. Cytarabine is the antimetabolite backbone and is dosed at standard 100-200 mg/m2/day continuous infusion in induction (high-dose cytarabine is reserved mainly for consolidation). The '7' in 7+3 always refers to the cytarabine days.

BOARD TRAP

In 7+3 the 7 days is cytarabine and the 3 days is the anthracycline — reversing them is a classic distractor. High-dose cytarabine belongs to consolidation, not standard induction.

3. Cytarabine = antimetabolite

WHAT YOU SEE

A label reading CYTARABINE ANTIMETABOLITE on the blue pipe — the tag marks Ara-C as the pyrimidine-analog backbone halting S-phase DNA synthesis.

WHAT IT ENCODES

Cytarabine is a pyrimidine (cytidine) analog antimetabolite: after intracellular phosphorylation to Ara-CTP it incorporates into DNA and inhibits DNA polymerase, halting S-phase synthesis. Its signature high-dose toxicities are cerebellar dysfunction (ataxia, nystagmus — check cerebellar exam before each dose and hold if abnormal), conjunctivitis (give prophylactic steroid eye drops), and chemical pancreatitis. It is the single most important drug in AML therapy.

BOARD TRAP

High-dose cytarabine neurotoxicity is cerebellar (ataxia/dysarthria), not peripheral neuropathy — and is an indication to HOLD the drug. Prophylactic steroid eye drops are required to prevent the cytarabine keratoconjunctivitis.

4. Anthracycline 3 days

WHAT YOU SEE

A red ANTHRACYCLINE pipeline feeding 3 dose drums — the red 3-dose line is the day 1-3 anthracycline of 7+3.

WHAT IT ENCODES

The '3' of 7+3 is an anthracycline given on days 1-3, most commonly daunorubicin or idarubicin. Anthracyclines provide the cytotoxic punch through topoisomerase II inhibition and free-radical DNA damage. Their dose-limiting long-term toxicity is cumulative, dose-dependent cardiomyopathy/heart failure, so a baseline echocardiogram (LVEF) is obtained before induction.

BOARD TRAP

Anthracycline cardiotoxicity is cumulative and dose-dependent (irreversible dilated cardiomyopathy); check baseline LVEF. The acute red urine after daunorubicin/idarubicin is harmless and must not be mistaken for hematuria.

5. Dauno 60 / Ida 12

WHAT YOU SEE

Dose tags reading DAUNO 60 and IDA 12 above the red pipeline — the labeled doses are standard daunorubicin 60-90 and idarubicin 12 mg/m2.

WHAT IT ENCODES

Standard induction doses are daunorubicin 60-90 mg/m2/day or idarubicin 12 mg/m2/day on days 1-3. Daunorubicin 90 mg/m2 (intensified) improves outcomes in younger fit patients versus 45 mg/m2. CPX-351 (liposomal daunorubicin + cytarabine) is the preferred induction for therapy-related AML and AML with myelodysplasia-related changes in older adults.

BOARD TRAP

For secondary/therapy-related AML and AML-MRC in older adults, CPX-351 (liposomal dauno/cytarabine) is preferred over conventional 7+3 — a frequently tested substitution.

6. Topo II / DNA crushed

WHAT YOU SEE

A TOPO II gauge over a strand of crushed/snapped DNA — the shattered helix is anthracycline topoisomerase-II inhibition causing double-strand breaks.

WHAT IT ENCODES

Anthracyclines act primarily by inhibiting topoisomerase II, trapping the enzyme-DNA cleavage complex and generating double-strand DNA breaks; they also intercalate DNA and generate reactive oxygen species. This same topo-II mechanism is why prior etoposide/anthracycline exposure causes short-latency therapy-related AML with KMT2A (11q23) rearrangements.

BOARD TRAP

Topoisomerase II inhibition is the anthracycline mechanism — and prior topo-II exposure later causes therapy-related AML with KMT2A/11q23 rearrangement and short (1-3 yr) latency, distinct from alkylator-induced t-AML.

7. FLT3 TKI add-on

WHAT YOU SEE

An orange FLT3 TKI tank plumbed into the line — the add-on tank is midostaurin bolted onto 7+3 for FLT3-mutated AML.

WHAT IT ENCODES

If AML is FLT3-mutated (ITD or TKD), add the FLT3 inhibitor midostaurin to 7+3 induction and consolidation — this improved overall survival in the RATIFY trial. For relapsed/refractory FLT3-ITD AML, gilteritinib is the agent of choice. FLT3-ITD carries adverse prognostic weight, so these patients are generally referred for allo-HCT in CR1.

BOARD TRAP

Midostaurin is added to 7+3 up front for FLT3-mutated AML (induction/consolidation); gilteritinib is for relapsed/refractory FLT3 disease. FLT3-ITD-positive patients should proceed to allo-HCT in CR1.

8. HMA + venetoclax (unfit)

WHAT YOU SEE

Two chairs labeled HMA (AZA/DEC) and VEN (BCL2) with a frail elderly patient — the geriatric station is the azacitidine+venetoclax regimen for unfit patients.

WHAT IT ENCODES

Older (typically ≥ 75 yr) or unfit patients who cannot tolerate intensive 7+3 receive a hypomethylating agent (azacitidine or decitabine) plus the BCL-2 inhibitor venetoclax — the VIALE-A regimen, which roughly tripled survival versus azacitidine alone. Venetoclax requires careful ramp-up with tumor lysis prophylaxis. IDH inhibitors (ivosidenib for IDH1, enasidenib for IDH2) are alternatives in mutated, unfit patients.

BOARD TRAP

Frail/elderly or comorbid patients → azacitidine/decitabine + venetoclax, NOT intensive 7+3. Venetoclax needs a ramp-up dose schedule with TLS prophylaxis; starting at full dose can precipitate fatal tumor lysis.

9. <5% blasts = CR

WHAT YOU SEE

A control-panel dial reading <5% in the green — the low blast gauge is the <5% marrow-blast threshold defining complete remission.

WHAT IT ENCODES

Complete remission requires <5% blasts in a marrow with adequate cellularity, no Auer rods, no extramedullary disease, AND peripheral count recovery (ANC ≥ 1000 , platelets $\geq 100k$). CR with incomplete count recovery (CRi) and measurable residual disease (MRD) positivity by flow/PCR predict relapse, so MRD status increasingly guides whether to escalate to transplant.

BOARD TRAP

CR demands BOTH <5% marrow blasts AND peripheral count recovery — <5% blasts without count recovery is only CRi. MRD positivity in apparent CR still portends relapse and may change management.

10. ANC / PLT recovery

WHAT YOU SEE

Indicator lights reading ANC >1000 and PLT >100k on the panel — the green recovery lamps are the count thresholds required (with <5% blasts) for true CR.

WHAT IT ENCODES

Hematologic recovery thresholds defining true CR are ANC >1000/ μ L and platelets >100,000/ μ L. A day-14 marrow is often checked to confirm blast clearance; persistent disease may prompt re-induction. Expect a profound 3-4 week aplastic nadir after 7+3 during which neutropenic fever and bleeding risk are highest.

BOARD TRAP

The CR count thresholds are ANC >1000 and platelets >100k — values short of these (e.g., persistent thrombocytopenia) downgrade the response to CRi, not CR.

11. Transfusion support

WHAT YOU SEE

Platelet and packed-red-cell bags at the supportive-care station — the blood products are the prophylactic transfusion support during induction aplasia.

WHAT IT ENCODES

During the induction aplasia, supportive care is as important as the chemo: transfuse platelets prophylactically (<10k, or higher if febrile/bleeding) and RBCs for symptomatic anemia, use leukoreduced and irradiated products to prevent CMV transmission and transfusion-associated GVHD, and give broad-spectrum antibacterial/antifungal/antiviral prophylaxis plus prompt empiric antibiotics for neutropenic fever.

BOARD TRAP

Blood products for AML/transplant candidates must be irradiated (prevents TA-GVHD) and leukoreduced (CMV risk, alloimmunization). Neutropenic fever is a medical emergency requiring empiric broad-spectrum antibiotics within an hour, before cultures result.

12. Allo / rasburicase

WHAT YOU SEE

Icons reading ALLO and RASB at the supportive station — the allopurinol/rasburicase pair is tumor-lysis prophylaxis started with induction.

WHAT IT ENCODES

Tumor lysis syndrome prophylaxis starts before/with induction: aggressive IV hydration plus allopurinol (xanthine oxidase inhibitor, prevents new urate formation) for lower-risk disease, and rasburicase (recombinant urate oxidase, degrades existing uric acid) for high tumor burden/hyperuricemia. Monitor for the TLS tetrad — hyperkalemia, hyperphosphatemia, hyperuricemia, hypocalcemia — with rising creatinine.

BOARD TRAP

Rasburicase, not allopurinol, is used when uric acid is already high (it breaks down existing urate) and is contraindicated in G6PD deficiency (causes hemolysis/methemoglobinemia). Allopurinol only prevents new urate and does not lower an already-elevated level.